SNAP Authorization and Local Retailer Participation*

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Abstract

We study how a major retailer's adoption of the Supplemental Nutrition Assistance Program (SNAP) influences the authorization decisions of other retailers in local markets. Exploiting Dollar General's chain-wide SNAP adoption in 2004 and using a difference-in-differences design, we find that the number of other SNAP-authorized grocery stores increases by 15 percent, with event study estimates showing that the effects persist and grow over time. We find evidence for three mechanisms. First, individual SNAP enrollment increases following Dollar General's adoption, expanding the customer base and making SNAP authorization more attractive to other retailers. Second, spillovers are larger in concentrated markets, consistent with strategic interdependence where firms face stronger incentives to respond to rivals' SNAP adoption. Third, effects are stronger in counties with low initial SNAP coverage, where Dollar General's adoption may reduce uncertainty about program viability or signal previously unmet demand to other retailers. These findings suggest that complementarities in retailers' participation decisions can extend program access beyond directly treated firms and indicate that private market dynamics play an important role in the delivery of public assistance programs.

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1 Introduction

Public assistance programs in the United States often operate at the intersection of government policy and private market participation (Currie and Gahvari, 2008). When entitlement programs rely on private firms for service delivery rather than direct government provision, program effectiveness becomes closely linked to the strategic behavior of market participants (Duggan, 2004). This institutional structure creates a critical but often underexamined channel through which policy design influences outcomes. Firms respond to program parameters, financial incentives, and competitive dynamics in ways that can generate multiplier effects or introduce constraints across local markets (Hendren and Sprung-Keyser, 2020). Yet traditional policy evaluations tend to emphasize benefit levels, eligibility, and recipient responses, often overlooking these supply-side mechanisms.

We examine how retailers' strategic participation decisions influence the market coverage of the Supplemental Nutrition Assistance Program (SNAP), focusing on whether Dollar General's SNAP adoption induced broader participation among other grocery stores in the same local market. When a retailer becomes SNAP-authorized, it generates multiple spillover mechanisms that can either reinforce or counteract program expansion. On the one hand, participation creates positive spillovers through two channels: market expansion effects arise when improved household access increases SNAP enrollment, enlarging the customer base and strengthening incentives for additional retailers to join; simultaneously, informational spillovers reduce uncertainty as early adopters demonstrate program viability and reveal operational requirements, lowering entry barriers for nearby retailers. On the other hand, participation generates negative competitive spillovers by redistributing customer traffic away from existing retailers, potentially rendering SNAP unprofitable for those bearing recurring fixed costs such as administrative compliance, specialized equipment maintenance, and staff training. The net effect of these countervailing forces—whether positive spillovers dominate to amplify program reach or negative spillovers prevail to concentrate access—determines equilibrium participation patterns and ultimately influences the program's effectiveness in reaching intended beneficiaries.

To identify retailer spillovers, we exploit a plausibly exogenous shock to local SNAP retail environments: Dollar General's chain-wide adoption of SNAP in 2004. Unlike typical authorization decisions that reflect individual store characteristics or coincide with store openings, Dollar General's decision was centralized and implemented uniformly across its existing store network. In a single year, more than 6,300 stores became SNAP-authorized. This large-scale, corporate-driven rollout creates a unique natural experiment to examine how the entry of a major retailer into SNAP influences the participation decisions of unaffiliated grocery stores operating in the same markets.

We construct a county-level panel spanning 2000-2008 using USDA's Historical SNAP Retailer Locator data, which provides precise information on retailer authorization timing, location, and characteristics. We supplement this with establishment-level data from the National Establishment Time Series (NETS) to measure total grocery store counts and market concentration. This comprehensive dataset allows us to track changes in both the number and share of SNAP-authorized retailers across geographic markets.

Using a difference-in-differences design, we compare counties with existing Dollar General stores that operated without SNAP authorization from 2000 to 2003 and then transitioned to become SNAP-authorized from 2004 to 2008, against counties without any SNAP-authorized Dollar General stores during our study period. This approach isolates the impact of Dollar General stores becoming SNAP-authorized, distinct from the effects of new store entry. We find that Dollar General's adoption of SNAP led to significant increases in program participation among other grocery retailers. Our baseline estimates indicate that the number of SNAP-authorized grocery stores increased by an average of 15 percent following Dollar General's transition. Event-study estimates suggest this effect reached nearly 30 percent by the fourth year after adoption. The share of grocery stores accepting SNAP increased by more than 6 percentage points. Using population-adjusted measures, we find that the number of SNAP-authorized stores per 1,000 residents increased by 0.028, and the number per 1,000 low-income residents increased by 0.163. We show that these effects are not driven by overall expansion in the local food retail sector, as the number of non-SNAP-authorized grocery stores remained unchanged.

To probe the underlying mechanisms, we examine the three spillover channels through which Dollar General's SNAP adoption could positively influence broader retailer authorization. First, we test for market expansion effects by examining changes in overall SNAP take-up. We find that the chain-wide adoption led to a statistically significant increase in individual enrollment in SNAP. This expansion of the SNAP customer base indicates that improved access through a major retailer drew new households into the program rather than merely reallocating existing participants among retailers, thereby enhancing the incentives for other retailers to seek SNAP authorization. Second, we examine whether the observed retailer spillovers reflect follow-on behavior influenced by local market structure. In concentrated markets, Dollar General's SNAP adoption may pose a more salient competitive threat, increasing rivals' incentives to adopt SNAP even when firms are differentiated. We find that the increase in SNAP-authorized stores is significantly larger in counties with higher initial concentration. This pattern aligns with theories of competitive imitation and evidence that follow-on entry is more pronounced in markets characterized by greater strategic interdependence, where rival actions are more visible and the pressure to respond is stronger (Lieberman and Asaba, 2006). Third, we examine whether spillover effects vary with initial SNAP retail coverage. We find that these effects are significantly larger in counties with lower initial retailer participation. This pattern suggests that Dollar General's SNAP adoption has stronger spillover effects in underserved markets, possibly due to informational spillovers that reduce uncertainty about program viability or to demand revelation that highlights previously unmet customer needs.

These mechanism findings are further supported by heterogeneity in spillover intensity across county demographic composition. We observe larger effects in counties with higher shares of children and Black residents—populations that historically exhibit higher SNAP enrollment rates and greater unmet demand for food retail access. These demographic patterns reinforce the market expansion mechanism, suggesting that Dollar General's adoption generated the largest spillovers in areas where the economic returns to SNAP authorization were most favorable due to substantial latent demand. Conversely, weaker responses in counties with larger senior populations align with limited expected takeup among older adults, consistent with reduced retailer incentives to participate in markets with lower potential SNAP utilization. Together, these results provide convergent evidence for a demand-driven mechanism, where market structure, initial coverage, and local demographic context jointly influence the magnitude of retailer spillovers.

Our event study estimates confirm the validity of the research design, showing no evidence of differential pre-trends between treated and control counties while revealing a steady increase in outcomes following treatment. The effects are robust to alternative specifications, including estimation at finer geographic levels, an extended sample period, and alternative control group definitions. To address potential violations of parallel trends, we also implement the Honest DiD framework of Rambachan and Roth (2023); our results remain statistically significant under both Relative Magnitude and Smoothness restrictions, suggesting robustness to moderate and gradually evolving deviations from identifying assumptions.

We contribute to three strands of literature. First, we extend research on how private firms shape the delivery and reach of public transfer programs. While most SNAP studies emphasize recipient-side outcomes such as food security and nutritional intake (e.g., Hoynes and Schanzenbach, 2015; Bailey et al., 2024), a growing literature investigates supply-side responses. Recent evidence suggests these supply-side factors matter: Li and Zhao (2025) find that proximity to SNAP authorized retailers, particularly superstores and grocery stores, is associated with improved diet quality among recipients. Byrne et al. (2024) document that SNAP expansions during the Great Recession spurred participation by non-traditional food retailers, improving geographic access without major changes in prices or diet. Oh (2024) shows that the national rollout of EBT led to declines in SNAP-authorized store counts, driven by exits of small-format retailers who faced burdensome upfront technology adoption costs and lacked the SNAP transaction volume to justify continued participation. We build on this work by showing that a major retailer's adoption of SNAP can generate spillovers on unaffiliated stores through demand growth and strategic responses.

Second, we contribute to the literature on strategic complementarities and competitive dynamics in firm entry decisions. Theoretical work shows that participation choices can be strategically complementary when firms' payoffs exhibit increasing differences in their own and others' actions, for example, when one firm's entry can increase the incentive for others to enter (Milgrom and Roberts, 1990; Vives, 2005). However, empirical evidence of these dynamics in real markets remains limited due to identification challenges. We provide novel evidence of strategic complementarity by exploiting Dollar General's chain-wide SNAP adoption as an exogenous shock to competitive environments. Our findings show that spillover effects are significantly larger in concentrated markets, where competitive interdependence is heightened and firms face greater pressure to match rivals' service offerings. This pattern aligns with theoretical perspectives suggesting that the intensity of strategic interaction and imitation increases in more concentrated market structures, where firms closely monitor and respond to rivals' actions (Lieberman and Asaba, 2006). Additionally, we find that spillovers are largest in markets with sparse initial program coverage, a pattern consistent with information transmission and learning effects operating alongside competitive dynamics. These findings contribute to the understanding of how market conditions influence strategic interactions among firms and document patterns suggestive of strategic complementarity in public program participation decisions.

Third, we provide insights into the role of demographic composition in influencing safety net program effectiveness. While prior work documents spatial disparities in program access (Bitler and Haider, 2011; U.S. Department of Agriculture Economic Research Service, 2021), less attention has been paid to how population characteristics affect the potential for program expansion through private sector engagement. Our analysis shows that retailer participation responses vary with local demographic profiles, with larger effects observed in areas where SNAP enrollment rates are historically higher. This variation suggests that demographic context may be an important consideration for understanding how private sector participation affects program reach across communities. The results indicate that population characteristics could be relevant for policy makers considering geographic approaches to expanding safety net program coverage through retailer participation.

The remainder of the paper is structured as follows. Section 2 provides institutional background on SNAP and Dollar General's 2004 chain-wide adoption of the program. Section 3 describes the data sources and sample construction. Section 4 presents the empirical specification. Section 5 reports our main findings, and Section 6 concludes.

2 Background

SNAP (formerly known as food stamps) is the largest food assistance program in the United States, serving over 42 million individuals in fiscal year 2023, with annual expenditures of about \$115 billion (Center on Budget and Policy Priorities, 2023). The program aims to reduce food insecurity and improve nutritional outcomes by providing benefits to low-income households. A large body of evidence documents its effectiveness, linking program participation to reductions in food hardship, improved health and educational outcomes, and enhanced long-term economic mobility (Hoynes and Schanzenbach, 2015, 2020; Bailey et al., 2024).

Unlike direct cash transfers, SNAP operates through a partnership between the government and private food retailers. Benefits can only be redeemed at SNAP-authorized stores, making the geographic distribution and characteristics of these retailers crucial determinants of program accessibility. Between 1998 and 2004, approximately 45-58% of food retailers participated in SNAP (Oh, 2024). The Food and Nutrition Service (FNS), a federal agency under USDA, oversees the retailer authorization process. To participate in SNAP, stores must apply and either maintain a depth of stock across staple food categories or derive more than 50% of their sales from staple foods (U.S. Department of Agriculture, Food and Nutrition Service, 2023). FNS also monitors compliance with program rules, with penalties for violations ranging from warnings to disqualification from the program.

Retailer participation in SNAP has expanded significantly over the past two decades. Between 2000 and 2020, the number of SNAP authorized food retailers increased by 52%, rising from approximately 172,000 to more than 260,000 (Food and Nutrition Service, 2025). Much of this growth occurred during the Great Recession, when program enrollment surged in response to economic hardship. Yet despite the expansion, access remains uneven. Large grocery chains and mass merchandise retailers, while comprising a minority of SNAP authorized stores, account for the majority of redemptions (Center on Budget and Policy Priorities, 2024). These outlets typically offer a wider selection of goods at lower prices but are more commonly located in higher-income and urban areas. In contrast, many low income and rural communities lack such retailers (Ver Ploeg et al., 2009), limiting the effective reach of SNAP.

Before 2004, most Dollar General stores were not authorized to accept SNAP. That year marked a major shift in the program's retail landscape when Dollar General—the U.S. retailer with the largest number of physical store locations (ScrapeHero, 2025; Retail Dive, 2024)—implemented a centralized, corporate-level decision to authorize SNAP transactions across its national network. This strategic rollout brought over 6,300 stores into the program, accounting for 86% of the chain's 7,320 locations at the time (Dollar General, 2004). The decision represented one of the largest single-year expansions of SNAP-authorized retailers in the program's history, substantially increasing access points across the different areas in which Dollar General operated. Figure 1 shows the sharp increase in SNAP-authorized Dollar General stores that occurred in 2004, when the chain implemented its nationwide adoption of the program.

This large-scale adoption provides a unique opportunity to examine how the entry of a major retail chain into SNAP influences the participation decisions of unaffiliated grocery retailers. Unlike typical SNAP authorization decisions, which are often shaped by localized demand, store-level profitability, or market-specific competition, Dollar General's chain-wide implementation stemmed from corporate strategy independent of local market conditions. As a result, the 2004 rollout generated a sharp, widespread, and plausibly exogenous shock to SNAP retail availability across different geographic contexts.

3 Data

3.1 SNAP-Authorized Retailer Data

We obtain administrative data on SNAP-authorized retailers from the Historical SNAP Retailer Locator Data provided by the USDA Food and Nutrition Service (FNS). This data includes comprehensive details on retailer names, types, geographic coordinates, and the start and end dates of SNAP authorization. Our analysis focuses on the period from 2000 to 2008, avoiding both data quality concerns in earlier years and confounding effects from major SNAP program changes following the American Recovery and Reinvestment Act in 2009.

We identify SNAP-authorized Dollar Stores based on retailer names, and compile a comprehensive list of SNAP-authorized Dollar General store locations. The dataset categorizes retailers into 17 distinct store types, with dollar stores primarily classified under the "combination grocery/other" category.[1] For each county-year, we calculate both the number of SNAP-authorized Dollar General stores and the total number of SNAP-authorized grocery stores. To isolate responses of other food retailers, we exclude Dollar General stores from all grocery store outcome measures.

[1] The 17 store types defined by USDA FNS include: Small Grocery Store, Medium Grocery Store, Large Grocery Store, Combination Grocery/Other, Supermarket, and Super Store/Chain Store (e.g., big box and warehouse clubs); as well as Convenience Store, Direct Marketing Farmer, Delivery Route, Farmers' Market, Internet Retailer, Military Commissary, Non-Profit Food Buying Cooperative, and Specialty Food Stores focused on Bakery/Bread, Fruits/Vegetables, Meat/Poultry, or Seafood.

3.2 National Establishment Time Series Data

We supplement the SNAP-authorized retailer data with establishment-level data from the National Establishment Time Series (NETS) database to obtain annual county-level counts of Dollar General and all grocery stores. Dollar General stores are identified within NETS using the same retailer name-based approach applied to the Historical SNAP Retailer Locator Data.[2]

Because the SNAP Retailer Locator Data does not include Standard Industrial Classification (SIC) codes, we leverage the structured SIC coding in NETS to construct a consistent classification of grocery stores. Following Melendez et al. (2024), we adopt the same SIC codes used in their study to systematically identify grocery stores in each county-year.[3]

The NETS database also provides establishment-level sales data, which we use to calculate market concentration measures. We use this information to construct the baseline Grocery Store Herfindahl-Hirschman Index (HHI) for each county in 2000.

We combine the NETS grocery store counts with the SNAP retailer data to construct key measures of SNAP retail market conditions used in our analysis. While the definition of grocery stores may not perfectly align between NETS and the SNAP Retailer Locator Data, any definitional discrepancies are unlikely to vary systematically across counties or over time, minimizing potential differential measurement error that could bias our DiD estimates.

3.3 County Characteristics and Food Retail Market Variables

County level demographic, economic, and program participation data are compiled from several sources. Demographic measures, including total population, age distribution, and racial composition, are collected from the US Census Bureau. Economic indicators such as unemployment rate, poverty rate, and median household income come from the Census Bu-

[2]We validate the total number of Dollar General stores identified in NETS against official store counts reported in Dollar General's annual reports. Our NETS-based counts closely align with the official national totals.

[3]The specific SIC codes used are 54110000, 54119901, 54119902, 54119903, 54119904, and 54119905.

reau's Small Area Income and Poverty Estimates program. Data on gross domestic product are sourced from the US Bureau of Economic Analysis.[4] County level SNAP participation data are obtained from the FNS.[5]

Based on these data, we construct several derived measures to characterize baseline county conditions in the year 2000. Racial composition is computed as the percentage of each racial group in the total population. Per capita GDP is calculated as total county GDP divided by total population. All demographic and economic variables are measured in 2000 to capture baseline county conditions at the start of our 2000-2008 study period.

To characterize local food retail market conditions at baseline, we construct several key measures using data from the year 2000. The *SNAP Retailer Share* represents the percentage of grocery stores that are SNAP-authorized, calculated by dividing the number of SNAP-authorized grocery stores by the total number of grocery stores from NETS. The *SNAP Retailer Density* captures the general availability of SNAP-accepting retailers, measured as the number of SNAP-authorized grocery stores per 1,000 residents. For a more targeted measure of SNAP retail availability for the population most likely to utilize benefits, we calculate the *SNAP Retailer Density (Low-Income)* as the number of SNAP-authorized grocery stores per 1,000 low-income residents.[6] The *Grocery Retailer HHI* sums the squared market shares of all grocery retailers in a county, with market shares based on store-level sales from NETS. Higher HHI values indicate greater market concentration and potentially less competition in local food retail markets.

Table 1 summarizes the key variables used in our analysis, including SNAP store availability, treatment assignment, and baseline county characteristics.

[4] We use real GDP measured in thousands of chained 2017 dollars.

[5] https://www.fns.usda.gov/pd/supplemental-nutrition-assistance-program-snap. Because not all states report enrollment at the county level, the sample size is smaller for analyses involving individual SNAP participation.

[6] "Low-income residents" refer to individuals living below the federal poverty line

4 Method

We implement a difference-in-differences (DiD) design using counties as the unit of analysis to isolate the causal effect of Dollar General's SNAP adoption on other retailers' participation decisions. Our research design capitalizes on the variation created by Dollar General's corporate-level policy shift while carefully defining treatment and control groups to address potential confounding factors.

The treatment group comprises counties that satisfy all of the following conditions: (1) at least one Dollar General store was present in the county each year from 2000 to 2008; (2) no Dollar General store in the county was SNAP-authorized during 2000–2003; and (3) from 2004 onward through 2008, the county had at least one SNAP-authorized Dollar General store each year. The control group consists of counties where no SNAP-authorized Dollar General store existed at any point during our study period. This approach isolates the effect of a store transitioning into SNAP authorization rather than capturing the effect of Dollar General's entry into a market, which could independently affect local retail dynamics.[7]

Our baseline specification is as follows:

$$Y_{it} = \alpha + \beta(\mathit{Treated}_i \times \mathit{Post}_t) + (X_i^{pre} \times f(t))'\boldsymbol{\gamma} + \mu_i + \eta_t + \varepsilon_{it}, \quad (1)$$

where i and t denote county and year, respectively. Y_{it} represents an outcome of interest, with the primary outcome defined as the log(count + 1) of SNAP-authorized grocery stores. We also examine additional outcomes, including the percentage of grocery stores accepting SNAP, the number of SNAP-authorized grocery stores per 1,000 residents, and the number of SNAP-authorized grocery stores per 1,000 low-income residents. Dollar General stores are excluded from the construction of all count-based outcome measures throughout the paper. $\mathit{Treated}_i$ equals 1 if county i belongs to the treatment group, and 0 otherwise; Post_t equals

[7] We interpret the estimated treatment effect as reflecting the market impact of Dollar General's transition into a SNAP-authorized food retailer, including any associated operational adjustments, such as expanded perishable food offerings.

1 if $t \geq 2004$, and 0 otherwise. The vector of pre-treatment control variables, X_i^{pre}, comprises county-level characteristics measured in 2000, the first year of the sample, including unemployment rate, poverty rate, log median household income, and log per capita GDP. To flexibly account for potential differential trends across counties, we follow Gentzkow (2006) and Li et al. (2016) by interacting X_i^{pre} with $f(t)$, where $f(t)$ denotes a set of first-, second-, and third-order polynomial functions of time.[8,9] We also include county fixed effects μ_i and year fixed effects η_t, and cluster standard errors at the county level. The coefficient of interest, β, captures the change in local grocery store SNAP participation before and after Dollar General's SNAP adoption for treatment counties relative to control counties.

To explore the dynamic effects of Dollar General's SNAP authorization and test for parallel trends, we estimate the following specification:

$$Y_{it} = \alpha + \sum_{k=-4}^{4} \beta_k D_{i,k} + (X_i^{pre} \times f(t))'\boldsymbol{\gamma} + \mu_i + \eta_t + \varepsilon_{it}, \tag{2}$$

where $D_{i,k}$ is a set of indicator variables equal to one if county i is k years relative to Dollar General's chain-wide SNAP adoption in 2004, and zero otherwise. The omitted category is $k = -1$, corresponding to the year immediately preceding Dollar General's chain-wide SNAP adoption. The coefficients of interest, β_k, can be interpreted as the average difference in SNAP participation outcomes between treatment and control counties at event time k, relative to event time $k = -1$. This specification traces the dynamic evolution of treatment effects over time and provides a diagnostic for the parallel trends assumption: if pre-treatment coefficients are small in magnitude and statistically indistinguishable from

[8] As a robustness check, we also estimate a specification in which each baseline control variable measured in 2000 is interacted with a linear time trend (i.e., a first-order polynomial), allowing counties to follow different trajectories based on initial conditions. Results are reported in the Appendix and remain consistent with our main estimates.

[9] In addition, we estimate a specification that interacts pre-treatment control variables with year dummies:

$$Y_{it} = \alpha + \beta(\mathit{Treated}_i \times \mathit{Post}_t) + (X_i^{pre} \times \mathit{YearDummy}_t)'\boldsymbol{\gamma} + \mu_i + \eta_t + \varepsilon_{it}$$

where definitions match Equation (1), except that $\mathit{YearDummy}_t$ denotes a set of year-specific dummy variables capturing year-by-year variations.

zero, this suggests that treatment and control counties followed similar trends prior to Dollar General's SNAP adoption in 2004.

To examine heterogeneity in treatment effects across regions with different pre-treatment characteristics, we extend the specification:

$$
\begin{aligned}
Y_{it} = \alpha &+ \beta(\mathit{Treated}_i \times \mathit{Post}_t) \\
&+ \theta(\mathit{Treated}_i \times \mathit{Post}_t \times C_i^{pre}) \\
&+ (X_{it}^{pre} \times f(t))'\boldsymbol{\gamma} + \mu_i + \eta_t + \varepsilon_{it}
\end{aligned} \tag{3}
$$

where C_i^{pre} denotes county-level characteristics measured in 2000. The coefficient θ captures how the treatment effect varies according to regional characteristics, allowing us to examine heterogeneous responses to Dollar General's SNAP authorization.

Finally, to estimate heterogeneous dynamic effects of SNAP authorization over time, we extend the event-study model by interacting event-time indicators with discretized pre-treatment characteristics, following Alsan and Wanamaker (2018) and Liu et al. (2024):

$$
\begin{aligned}
Y_{it} = \alpha &+ \sum_{k=-4}^{4} \beta_k (D_{i,k} \times C_i^{disc}) \\
&+ (X_i^{pre} \times f(t))'\boldsymbol{\gamma} + \mu_i + \eta_t + \varepsilon_{it}
\end{aligned} \tag{4}
$$

where C_i^{disc} denotes a discretized version of the continuous pre-treatment characteristic C_{it}^{pre}. To construct C_{it}^{disc}, we calculate the empirical distribution of C_{it}^{pre} across all counties measured in 2000 and identify the 33rd and 66th percentile cutoffs. Counties are then classified into three mutually exclusive groups based on these cutoffs: (1) *Low* group if $C_{it}^{pre} < p_{33}$; (2) *Middle* group if $p_{33} \leq C_{it}^{pre} < p_{66}$; and (3) *High* group if $C_{it}^{pre} \geq p_{66}$. This tercile-based classification enables us to assess potential nonlinear heterogeneity in treatment effects while maintaining statistical power across subgroups.

5 Results

5.1 Baseline Effects on SNAP Retailer Authorization

Table 2 presents the difference-in-differences estimates of the impact of Dollar General's SNAP adoption on local SNAP retailer participation.

Looking first at the number of SNAP-authorized retailers (columns 1-5), we find that Dollar General's entry into the SNAP program increases the number of SNAP-authorized grocery stores in the same county by approximately 13.1 percent in our baseline specification with only county and year fixed effects (column 1). This effect increases slightly to 14.9 percent when we account for differential trends across counties with varying economic conditions by adding interactions between control variables and year dummies (column 2).

To ensure our findings are not sensitive to model specification, we progressively add more flexible time trends in columns 3-5. Column (3) includes linear time trends interacted with county controls, column (4) adds quadratic interactions, and our preferred specification in column (5) incorporates cubic interactions. The estimated impact remains stable at approximately 15 percent across these specifications, suggesting that our results are robust to different functional forms of time-varying county characteristics.

Using our preferred specification with the full set of controls and polynomial time trends, we examine alternative measures of SNAP retail participation in columns (6) through (8). For the percentage of grocery stores accepting SNAP (column 6), we find that Dollar General's SNAP adoption increases the SNAP authorization rate among local grocery retailers by 6.38 percentage points. Column (7) examines SNAP retailer density, measured as SNAP-authorized grocery stores per 1,000 residents, and shows an increase of 0.028 stores per 1,000 residents following Dollar General's SNAP adoption. Column (8) presents an alternative normalization that focuses specifically on accessibility for low-income populations, measured as SNAP-authorized grocery stores per 1,000 residents below the poverty line. This measure shows an effect of 0.163 additional SNAP-authorized stores per 1,000 low-income residents.

Both measures suggest the positive impact of Dollar General's SNAP adoption on SNAP retailer accessibility. In general, all specifications consistently imply that Dollar General's adoption of SNAP led to a positive response among other grocery retailers.

5.2 Dynamic Effects on SNAP Retailer Authorization

In this subsection, we examine the dynamic effects of Dollar General's SNAP adoption on local food retail markets and assess the parallel trends assumption underlying our difference-in-differences design. The corresponding estimates are presented in Figure 2, which plots results from the specification in Equation 2 across four outcomes reported in Table 2: log number of SNAP-authorized grocery stores (Panel A), SNAP retailer share (Panel B), SNAP retailer density (Panel C), and SNAP retailer density for low-income populations (Panel D).

In Panel A, the estimates are close to zero and statistically insignificant in the pre-treatment period, supporting the parallel trends assumption. Following Dollar General's SNAP adoption, we observe positive and statistically significant impacts that grow steadily over time. By year 4, SNAP-authorized grocery stores increase by approximately 30 percent relative to the pre-adoption period.

Panel B shows a similar pattern for SNAP retailer share. The pre-adoption estimates are small and statistically indistinguishable from zero, supporting the parallel trends assumption. Following SNAP adoption by Dollar General, the estimated effect grows steadily over time, reaching about 5 percentage points by year 2 and continuing to rise to about 10 percentage points by year 4.

In Panel C, SNAP retailer density displays the same pattern of null pre-treatment effects followed by growing positive impacts. The estimates before Dollar General's SNAP adoption hover around zero with narrow confidence intervals, providing further evidence that treatment and control counties followed parallel trajectories. After Dollar General's entry into SNAP, we observe a gradual increase in retail accessibility, reaching approximately 0.04 additional SNAP-authorized grocery stores per 1,000 residents by year 4.

Panel D shows the results for SNAP retailer density among low-income populations. The pre-treatment estimates fluctuate around zero, with wide confidence intervals and a modestly positive estimate in one year before the event, indicating that the parallel trends assumption may be somewhat weaker for this outcome. Following Dollar General's SNAP adoption, the effect increases sharply, reaching about 0.3 additional SNAP-authorized grocery stores per 1,000 low-income residents by year 1. The effect continues to rise and stabilizes between 0.3 and 0.4 through years 2 to 4.

Together, these dynamic estimates complement the baseline results in Table 2. They confirm the absence of pre-treatment trends and show that the effects of Dollar General's SNAP adoption grow over time, suggesting a sustained and expanding response among other grocery retailers.

5.3 Spillover Mechanisms

To provide insights into the underlying mechanisms driving the observed positive retailer spillovers, we examine the three channels through which Dollar General's SNAP adoption could influence local retailer SNAP participation. Table 3 presents evidence on each mechanism, while Figure 3 displays corresponding event study estimates.

We first test for market expansion effects by examining changes in overall SNAP takeup. Columns (1) and (2) of Table 3 suggest that Dollar General's chain-wide adoption led to statistically significant increases in individual SNAP enrollment. The number of enrolled SNAP individuals increases by 17.2 percent, while the enrollment rate, defined as SNAP enrollment per capita, increases by 1.85 percentage points. The top panels of Figure 3 present these effects dynamically, showing steady increases in both SNAP enrollment and enrollment rates following Dollar General's adoption. While the enrollment rate panel (right) exhibits flat pre-treatment estimates consistent with parallel trends, the enrollment level panel (left) shows some evidence of differential pre-trends, with treated counties experiencing lower enrollment growth in the years preceding Dollar General's SNAP adoption. However, the sharp upward shift following treatment in both panels, combined with the sustained positive effects, suggests that Dollar General's entry generated meaningful increases in individual SNAP enrollment. This growth in individual enrollment suggests that Dollar General's adoption expanded overall program utilization rather than simply reallocating existing participants among retailers.

We next examine whether the observed retailer spillovers reflect follow-on behavior amplified by local market structure. In concentrated markets, where fewer competitors operate and strategic interdependence is greater, unauthorized grocery stores may have stronger incentives to respond to a major retailer's SNAP adoption to avoid losing market share. Column (3) shows that the interaction between treatment and the grocery store Herfindahl-Hirschman Index (HHI) is positive and statistically significant, indicating that the increase in SNAP-authorized stores is significantly larger in counties with higher initial concentration. This pattern is consistent with theories of competitive imitation, which propose that firms may be more likely to mimic rivals' strategic moves in environments characterized by heightened strategic interdependence—conditions that often arise in concentrated markets where competitors' actions are more visible and perceived as more threatening to a firm's position (Lieberman and Asaba, 2006). The bottom-left panel of Figure 3 illustrates this pattern dynamically, showing that post-treatment increases in SNAP-authorized stores are larger in high-concentration counties. While pre-treatment trends are less stable in low-concentration areas, the overall divergence in post-treatment effects suggests that follow-on retailer responses are larger in markets where firms face greater strategic pressure to react to a rival's SNAP adoption.

We next examine whether the observed retailer spillovers reflect follow-on behavior amplified by local market structure. In markets with higher concentration, unauthorized grocery stores may face greater pressure to match a major competitor's service offerings to protect their customer base. Column (3) shows that the interaction between treatment and the grocery store Herfindahl-Hirschman Index (HHI) is positive and statistically significant, indicating that spillover effects are larger in counties with higher initial concentration. This finding supports theories of competitive imitation, which predict stronger rival responses in environments where strategic interdependence heightens the visibility and potential impact of competitors' actions (Lieberman and Asaba, 2006). The bottom-left panel of Figure 3 illustrates this pattern dynamically, showing that post-treatment increases in SNAP-authorized stores are larger in high-concentration counties. While pre-treatment trends are less stable in low-concentration areas, the overall divergence in post-treatment effects confirms that follow-on retailer responses are amplified where strategic interdependence is strongest.

Finally, we examine whether spillover effects vary with initial SNAP retail coverage. Column (4) shows a negative and statistically significant interaction between treatment and the initial SNAP retailer share, which suggests that spillover effects are larger in counties with lower initial coverage. The bottom-right panel of Figure 3 displays this pattern over time, showing that counties with limited initial SNAP coverage experienced greater increases in retailer authorization following Dollar General's adoption compared to counties where SNAP participation was already widespread among local retailers. This pattern is consistent with Dollar General's SNAP adoption having larger spillover effects in areas with lower initial coverage, potentially driven by informational spillovers, where Dollar General's adoption reduces uncertainty about program viability, or demand revelation, where its success signals previously unmet customer needs to other retailers.

The results point to multiple channels through which Dollar General's SNAP adoption influences broader retailer participation. Market expansion provides the demand-side foundation for spillovers by increasing program enrollment. In addition, the strength of retailer responses varies systematically across markets, with larger effects observed in more concentrated and less-covered areas. These patterns are consistent with mechanisms that amplify spillovers through strategic interdependence in concentrated markets and information revelation in underserved areas. In particular, the evidence aligns with strategic complementarity in SNAP participation decisions, where the value of program entry rises when neighboring retailers also participate.

5.4 Heterogeneous Effects by County Demographic Composition

The mechanism analysis shows that Dollar General's SNAP adoption increased individual SNAP enrollment, suggesting demand-driven market expansion, and that spillover effects were amplified in concentrated markets and areas with low initial SNAP coverage. To further explore how local context relates to these effects, we examine heterogeneity in retailer responses by county demographic composition.

Demographic characteristics shape both SNAP eligibility and participation rates, and may therefore influence the economic returns to program participation for local retailers. We focus on four population measures: the shares of children, older adults, non-White residents, and Black residents in each county (measured in 2000). These groups represent distinct enrollment patterns in SNAP and differing levels of structural access to food retailers. For instance, households with children have consistently higher SNAP enrollment rates, while eligible seniors participate at much lower rates due to stigma, complexity, and limited mobility (Center on Budget and Policy Priorities, 2024; National Council on Aging, 2022). Similarly, communities with higher shares of non-White and Black residents often face longstanding retail access barriers (Walker et al., 2010; Allcott et al., 2019).

Table 4 presents heterogeneous treatment effects of Dollar General's SNAP adoption by county demographic composition. Column (1) shows a positive and statistically significant interaction between treatment and the share of residents under age 18, indicating that counties with more children experience larger retailer spillovers. This pattern is consistent with higher SNAP take-up among families with children and is reflected in the top-left panel of Figure 4, where the post-treatment gap between high and low-child-share counties steadily widens, with no evidence of differential pre-trends.

Column (2) shows a negative and statistically significant interaction with the share of the population over age 65, suggesting that counties with more seniors experienced weaker spillovers. This result aligns with lower SNAP participation rates among older adults and limited marginal benefits for retailers in those markets. The corresponding event study (top-right panel of Figure 4) shows significantly smaller post-treatment effects in counties with larger senior populations. Both high- and low-senior-share groups exhibit similar pre-treatment trends, consistent with the parallel trends assumption.

In columns (3) and (4), we examine racial composition. Both the non-White share and the Black share are associated with significantly larger post-treatment effects, with the interaction terms positive and statistically significant. These estimates suggest that spillovers are larger in racially mixed counties, particularly those with larger Black populations. The bottom panels of Figure 4 reinforce this pattern, showing that treatment effects in high-share counties emerge after Dollar General's SNAP adoption and grow over time, with parallel trends prior to treatment.

5.5 Robustness Checks

We conduct a series of robustness checks to assess the sensitivity of our main result. Table 5 presents difference-in-differences estimates across four alternative specifications, and Figure 5 displays the corresponding event study estimates. To further assess the validity of our identification strategy, we also implement the Honest DiD sensitivity analysis developed by Rambachan and Roth (2023). This approach allows us to formally evaluate how much deviation from the parallel trends assumption would be required to render our estimated effects statistically insignificant. This is relevant in our setting, where SNAP authorization patterns may be influenced by slowly evolving demographic or market conditions that differ across counties.

We first examine whether the observed effects reflect overall growth in the grocery retail sector rather than a shift in SNAP participation. Using the number of non-SNAP-authorized grocery stores as the outcome, we find a statistically insignificant estimate, suggesting that our main results are not likely driven by broader store expansion. The event study in the top-left panel of Figure 5 supports this, showing flat estimates both before and after adoption.

We then re-estimate the baseline model at the census tract level to assess whether our results are driven by spatial aggregation. We find that Dollar General's SNAP adoption increases the number of SNAP-authorized grocery stores by 11.4 percent, which confirms that the effect persists at finer geographic resolution. The top-right panel of Figure 5 shows generally parallel trends prior to SNAP adoption, with only a minor deviation in one pre-treatment period, and a clear upward shift in SNAP participation afterward.

Next, we extend the sample period through 2019 to test for longer-term effects. While our main analysis focuses on the 2000–2008 period to minimize confounding from the Great Recession and subsequent changes in SNAP policy and retail market dynamics, the extended window allows us to assess the persistence of the observed effects. The estimated impact increases to 28.5 percent, which suggests that the effects grow over time. The bottom-left panel of Figure 5 shows a continued increase in SNAP-authorized grocery stores through year 15 after Dollar General's SNAP adoption, with no evidence of pre-treatment differences between treated and control counties.

We also test the sensitivity of our findings to the definition of the control group. In our main specification, control counties may include locations with Dollar General stores that were not yet SNAP-authorized. While these counties are untreated by construction, their retail environments may still differ systematically from those without any Dollar General presence. To address this concern, we exclude all counties that had any Dollar General stores, regardless of SNAP status, from the control group, ensuring a cleaner counterfactual. The estimated treatment effect remains positive and statistically significant at 14.9 percent. As shown in the bottom-right panel of Figure 5, the post-treatment increase in SNAP participation persists, and there are no parallel pre-trends.

Finally, we assess the sensitivity of our findings to potential violations of the parallel trends assumption using the Honest DiD framework proposed by Rambachan and Roth (2023). Figure 6 presents bounds on the average treatment effect across the post-treatment period (*post_1* to *post_4*) under restrictions on Relative Magnitude (RM) and Smoothness.[10] The RM restriction limits post-treatment deviations to no more than $Mbar$ times the largest change observed between adjacent pre-treatment periods. The left panel shows that the estimated effect remains statistically significant even when $Mbar = 0.5$, which indicates robustness to moderate violations in the relative size of pre- and post-trend slopes. The Smoothness restriction places a direct bound on the rate at which deviations can evolve over time by limiting changes in the slope of the violation. This restriction is especially relevant when potential departures from parallel trends are expected to be gradual. In the right panel, the estimate remains statistically significant for smoothness values up to $M = 0.0158$. Overall, the results suggest that the main findings are not sensitive to modest or slowly evolving violations of the parallel trends assumption.

6 Conclusion

Delivering SNAP benefits through private retailers is an important feature of the U.S. food assistance programs, yet little is known about how one firm's actions influence program access for others. This paper provides evidence that retailers' decisions to participate in SNAP create spillover effects that extend program reach beyond the firms directly involved. Exploiting Dollar General's chain-wide SNAP adoption in 2004 as a plausibly exogenous shock, we document an unintended but beneficial consequence of a large retailer's strategic choice: Local grocery stores increased their SNAP authorization by an average of 15 percent following Dollar General's transition, with event-study estimates showing effects approaching 30 percent by the fourth year. To our knowledge, this is the first paper to quantify retailer-to-retailer spillovers in SNAP participation with establishment-level data.

We show that the strength of retailer spillovers varies with local market conditions and

[10] In the Honest DiD framework, $Mbar$ governs how large the post-treatment violation of the parallel trends assumption can be relative to the largest observed change in the pre-treatment period. M controls the smoothness of the violation path by bounding changes in the slope of deviations across time. Lower values of $Mbar$ and M imply tighter restrictions on how much and how abruptly trends can diverge post-treatment.

demographic composition, consistent with a demand-driven mechanism. First, we find that Dollar General's SNAP adoption increased individual program enrollment, indicating that it expanded the overall market size rather than merely reallocating existing participants. This result supports market expansion as a mechanism behind the observed spillover effects. Second, we show that spillovers were larger in more concentrated retail markets, consistent with firms facing greater competitive pressure and stronger incentives to respond strategically to rivals' program participation. Third, we find that retailer spillovers were larger in areas with lower initial SNAP retailer coverage, potentially driven by informational spillovers that reduce uncertainty about program viability or demand revelation that signals previously unmet customer needs. Finally, we show that spillovers were larger in counties with higher shares of children and Black residents—populations with high SNAP eligibility and persistent barriers to food retail access—where the returns to program participation are likely greatest. These patterns align with established differences in SNAP take-up and retail availability, and point to stronger retailer incentives in areas of concentrated demand. In contrast, weaker responses in counties with larger senior populations reflect lower SNAP enrollment rates among older adults and diminished marginal benefits to store entry.

Our findings have several implications. First, our results suggest that targeting large retailers for SNAP participation could generate substantial indirect benefits through competitor responses, potentially offering a cost-effective approach to expanding program reach. Second, the heterogeneous effects indicate that policymakers should consider local market conditions and demographic characteristics when developing strategies to improve SNAP access. Event-study estimates show that these gains persist for multiple years, providing potentially long-term improvements in program delivery.

More broadly, our findings highlight the importance of viewing safety net programs like SNAP as embedded in local market systems, where private firms function as critical nodes in the delivery infrastructure. Rather than acting in isolation, firms respond to one another's participation decisions in ways that shape program coverage and accessibility. When public benefits are delivered through private channels, the strategic interdependence of market participants may influence how effectively policy interventions reach their intended beneficiaries.

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Tables

Table 1: Summary Statistics

Variable	N	Mean	SD
Number of SNAP-Authorized Grocery Stores	12,258	19.7	108.9
Treated × Post	12,258	0.21	0.41
SNAP-Authorized Grocery Stores per 1,000 People	12,231	0.25	0.32
SNAP-Authorized Grocery Stores per 1,000 Low-Income	12,256	2.05	2.77
Share of SNAP-Authorized Grocery Stores (%)	11,987	53.9	71.0
Number of Non-SNAP-Authorized Grocery Stores	12,231	24.1	85.1
SNAP Enrollment (thousands)	9,251	8.1	30.7
SNAP Enrollment Rate (%)	9,234	8.3	6.6
County Characteristics in 2000			
Total Population (thousands)	12,249	113.9	389.1
Unemployment Rate (%)	12,249	3.5	2.0
Poverty Rate (%)	12,249	9.9	5.3
Median Household Income (USD, thousands)	12,249	36.7	9.6
Per Capita GDP (USD, thousands)	12,060	39.9	33.9
Share Under 18 (%)	12,249	35.1	5.9
Share Over 65 (%)	12,249	14.7	4.6
Share of Non-White Population (%)	12,249	14.9	16.3
Share of Black Population (%)	12,222	6.1	11.5
Share of SNAP-Authorized Grocery Stores (%)	12,015	56.4	87.4
SNAP-Authorized Grocery Stores per 1,000 People	12,249	0.26	0.36
SNAP-Authorized Grocery Stores per 1,000 Low-Income	12,249	2.71	3.49
Grocery Retailer HHI	12,033	0.33	0.25

Notes: This table presents summary statistics for the main analysis sample, including the number of observations (N), mean, and standard deviation (SD) for each variable. The upper panel presents statistics for the 2000–2008 sample period. SNAP-authorized grocery stores are obtained from the Historical SNAP Retailer Locator Data, excluding SNAP-authorized Dollar General stores. Treated × Post is an indicator for treated counties in post-adoption years. Share of SNAP-Authorized Grocery Stores (%) is the percentage of grocery stores in a county that are SNAP-authorized, while SNAP-Authorized Grocery Stores per 1,000 People and per 1,000 Low-Income capture retailer density relative to the total population and the population living below the federal poverty line, respectively. Number of Non-SNAP-Authorized Grocery Stores is the difference between the total number of grocery stores and those authorized for SNAP. SNAP Enrollment refers to the number of enrolled SNAP recipients in a county, while SNAP Enrollment Rate is the share of the total county population enrolled in SNAP. The lower panel reports baseline county demographic and economic characteristics in 2000, with demographic data, including total population, age distribution, and racial composition, sourced from the U.S. Census Bureau; unemployment rate, poverty rate, and median household income from the SAIPE program; and real per capita GDP (in thousands of chained 2017 dollars) from the Bureau of Economic Analysis. Racial composition measures are calculated as population shares by group. Grocery Retailer HHI reflects the Herfindahl-Hirschman Index based on store-level sales from NETS, indicating market concentration within county food retail sectors.

Table 2: Impact of Dollar General's SNAP Adoption on SNAP Participation in Local Food Retail Markets

	Log Number of SNAP Retailer					SNAP Retailer Share	SNAP Retailer Density	SNAP Retailer Density (Low-Income)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated × Post	0.131***	0.149***	0.147***	0.147***	0.148***	6.380***	0.028***	0.163***
	(0.018)	(0.017)	(0.017)	(0.017)	(0.017)	(1.523)	(0.007)	(0.052)
County Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Controls × Year Dummies		Y						
Controls × Linear Trend			Y	Y	Y	Y	Y	Y
Controls × Quadratic Trend				Y	Y	Y	Y	Y
Controls × Cubic Trend					Y	Y	Y	Y
Observations	12,258	12,249	12,060	12,060	12,060	11,791	12,033	12,059

Notes: This table reports the DiD estimates of the impact of Dollar General's SNAP adoption on SNAP participation by other grocery stores in local markets. All SNAP-authorized Dollar General store counts are excluded from the outcome measures: log(count + 1) of SNAP-authorized grocery stores (columns 1–5); percentage of grocery stores accepting SNAP (column 6); SNAP-authorized grocery stores per 1,000 residents (column 7); and SNAP-authorized grocery stores per 1,000 residents below the poverty line (column 8). Control variables (measured in 2000) include unemployment rate, poverty rate, log median household income, and log per capita GDP. These controls are interacted with year dummies and time trends, which represent first-order (linear), second-order (quadratic), and third-order (cubic) polynomial functions of time. Standard errors (in parentheses) are clustered at the county level, with *, **, and *** indicating significance at 10%, 5%, and 1% levels, respectively.

Table 3: Mechanism for Dollar General's SNAP Adoption Effects

	(1) SNAP Enrollment	(2) SNAP Enrollment Rate	(3) Grocery Retailer HHI	(4) SNAP Retailer Share
Treated × Post	0.172***	1.847***	0.049*	0.238***
	(0.014)	(0.146)	(0.026)	(0.029)
Treated × Post × Grocery Retailer HHI			0.030**	
			(0.012)	
Treated × Post × SNAP Retailer Share				-0.0004***
				(0.0001)
County Fixed Effects	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y
Controls × Polynomial Trends	Y	Y	Y	Y
Observations	9,083	9,065	11,853	11,826

Notes: This table presents evidence on the mechanisms underlying retailer spillover effects following Dollar General's SNAP adoption. Columns (1) and (2) test for market expansion by estimating the effect on individual SNAP participation outcomes. The dependent variables are the log of the number of enrolled SNAP individuals and SNAP enrollment per capita, respectively. Columns (3) and (4) assess heterogeneity in the effect on the number of SNAP-authorized grocery stores (excluding SNAP-authorized Dollar General stores) based on initial county food retail market characteristics measured in 2000. Column (3) interacts the treatment with the Grocery Retailer HHI (Herfindahl-Hirschman Index), capturing variation by market concentration. Column (4) interacts with SNAP Retailer Share (the percentage of grocery stores that were SNAP-authorized in 2000), capturing variation by initial program coverage. All models include county and year fixed effects and control variables interacted with first- through third-order polynomial time trends. Standard errors (in parentheses) are clustered at the county level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 4: Heterogeneous Effects of Dollar General's SNAP Adoption by County Demographics

	(1)	(2)	(3)	(4)
Treated × Post	-0.2491***	0.4205***	0.0549**	0.0990***
	(0.0363)	(0.0650)	(0.0240)	(0.0212)
Treated × Post × Share Under 18	0.0016***			
	(0.0002)			
Treated × Post × Share Over 65		-0.0196***		
		(0.0044)		
Treated × Post × Share Non-White			0.0008***	
			(0.0002)	
Treated × Post × Share Black				0.0007**
				(0.0003)
County Fixed Effects	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y
Controls × Polynomial Trends	Y	Y	Y	Y
Observations	12,060	12,060	12,060	12,033

Notes: This table reports heterogeneous treatment effects of Dollar General's SNAP adoption by county demographics in 2000. The dependent variable is log(count + 1) of SNAP-authorized grocery stores (excluding SNAP-authorized Dollar General stores). Each column reports the interaction effect between Treated × Post and a demographic characteristic: share under 18, share over 65, share non-White, and share Black, measured in percentage points. All models include county and year fixed effects and control variables interacted with first-, second-, and third-order polynomial time trends. Standard errors (in parentheses) are clustered at the county level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 5: Robustness Checks of Dollar General's SNAP Adoption Effects

	Impact on Non-SNAP Grocery Stores (1)	Census Tract Model (2)	Extended Sample Period to 2019 (3)	Exclude Counties with Dollar General from Control Group (4)
Treated × Post	-0.013	0.114***	0.285***	0.149***
	(0.020)	(0.014)	(0.025)	(0.017)
County/Tract Fixed Effects	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y
Controls × Polynomial Trends	Y	Y	Y	Y
Observations	11,074	337,122	25,900	12,018

Notes: This table reports robustness checks of the baseline analysis. The dependent variable is log(count + 1) of non-SNAP-authorized grocery stores in column 1, and log(count + 1) of SNAP-authorized grocery stores (excluding SNAP-authorized Dollar General stores) in columns 2, 3, and 4. Column 1 uses non-SNAP-authorized grocery stores as the outcome. Column 2 re-estimates the baseline model at the census tract level. Column 3 extends the sample period through 2019. Column 4 excludes control counties containing Dollar General stores. All models include county (or tract) and year fixed effects and control variables interacted with first-, second-, and third-order polynomial time trends. Standard errors (in parentheses) are clustered at the corresponding level of analysis. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Figures

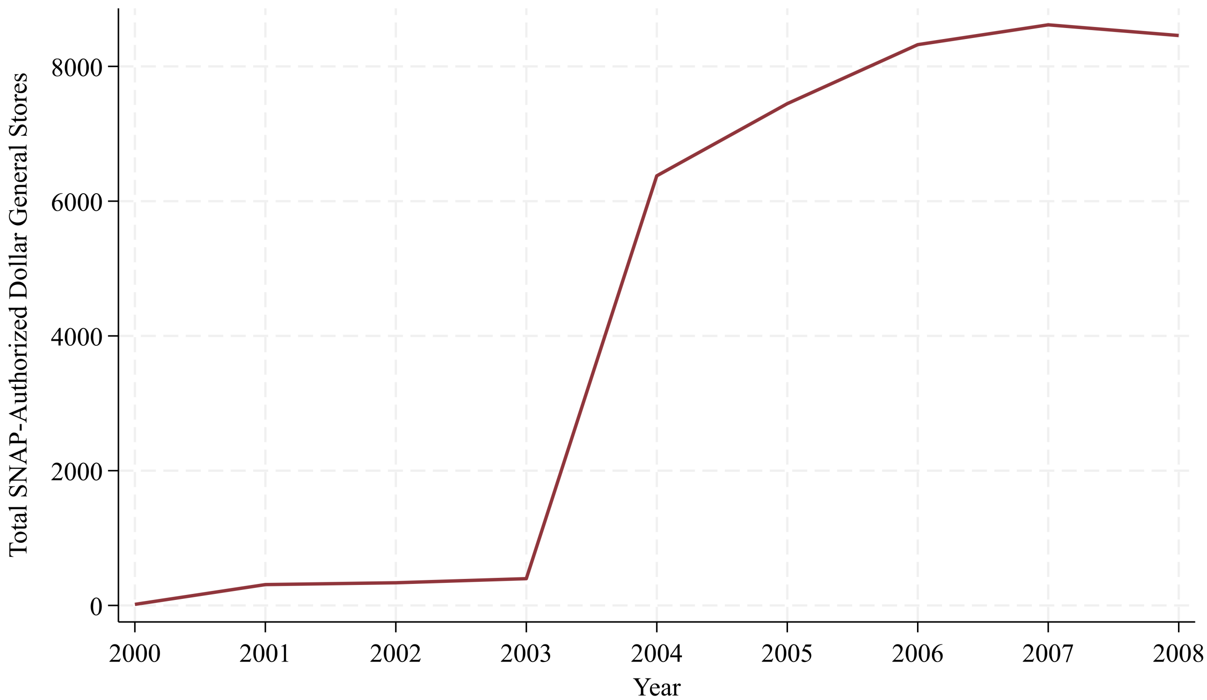

Figure 1: Number of SNAP-Authorized Dollar General Stores in the U.S., 2000–2008

Source: Historical SNAP Retailer Locator Data.

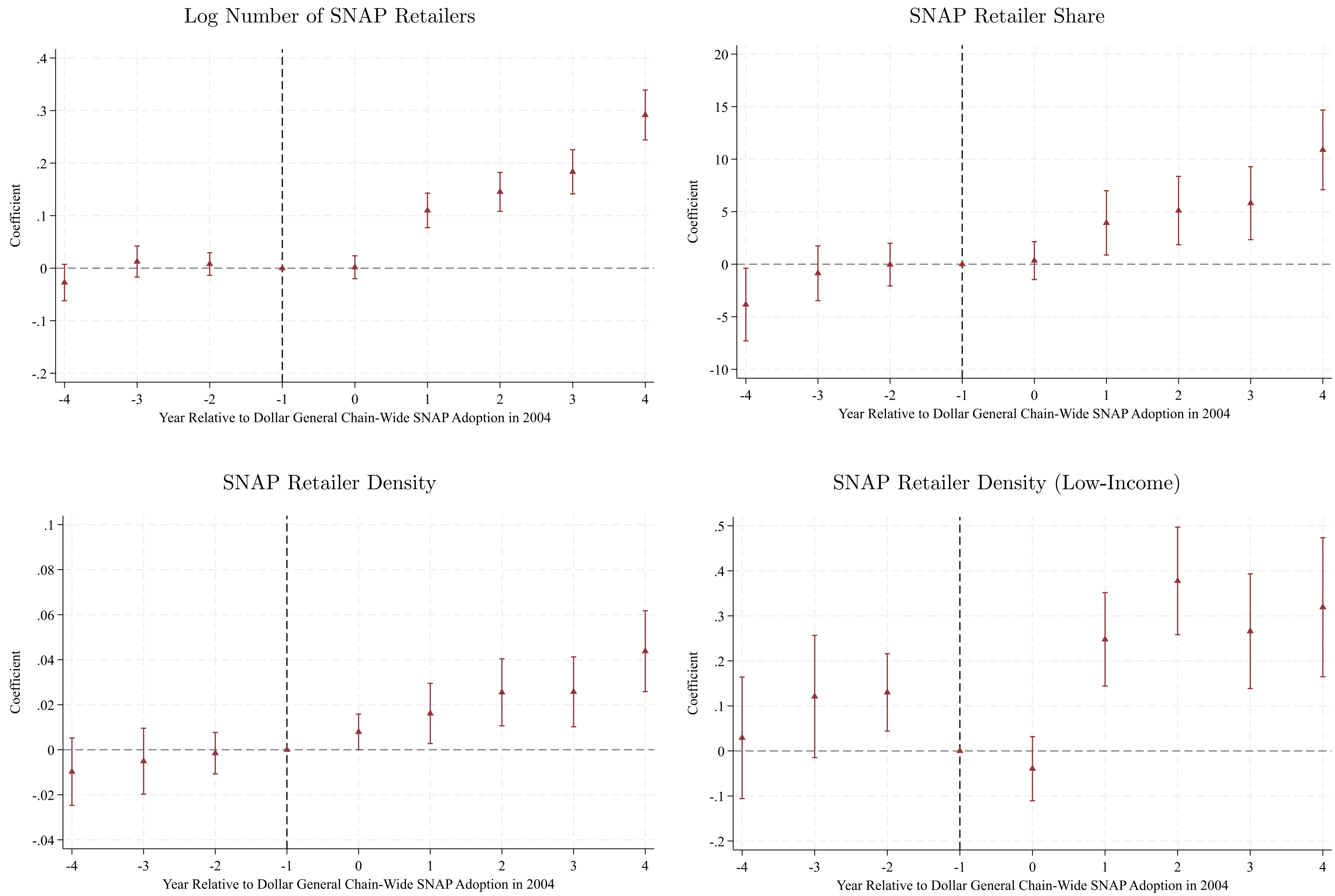

Figure 2: Event Study of Dollar General SNAP Adoption

Notes: This figure presents event study estimates of the effects of Dollar General's chain-wide SNAP adoption, based on the specification in Equation (2), across the four outcome variables reported in Table 2. The dependent variable is the log(count + 1) of SNAP-authorized grocery stores (excluding Dollar General). The x-axis denotes years relative to 2004, the year of Dollar General's chain-wide SNAP adoption. Each point represents the estimated coefficient for the corresponding period; bars represent 95% confidence intervals.

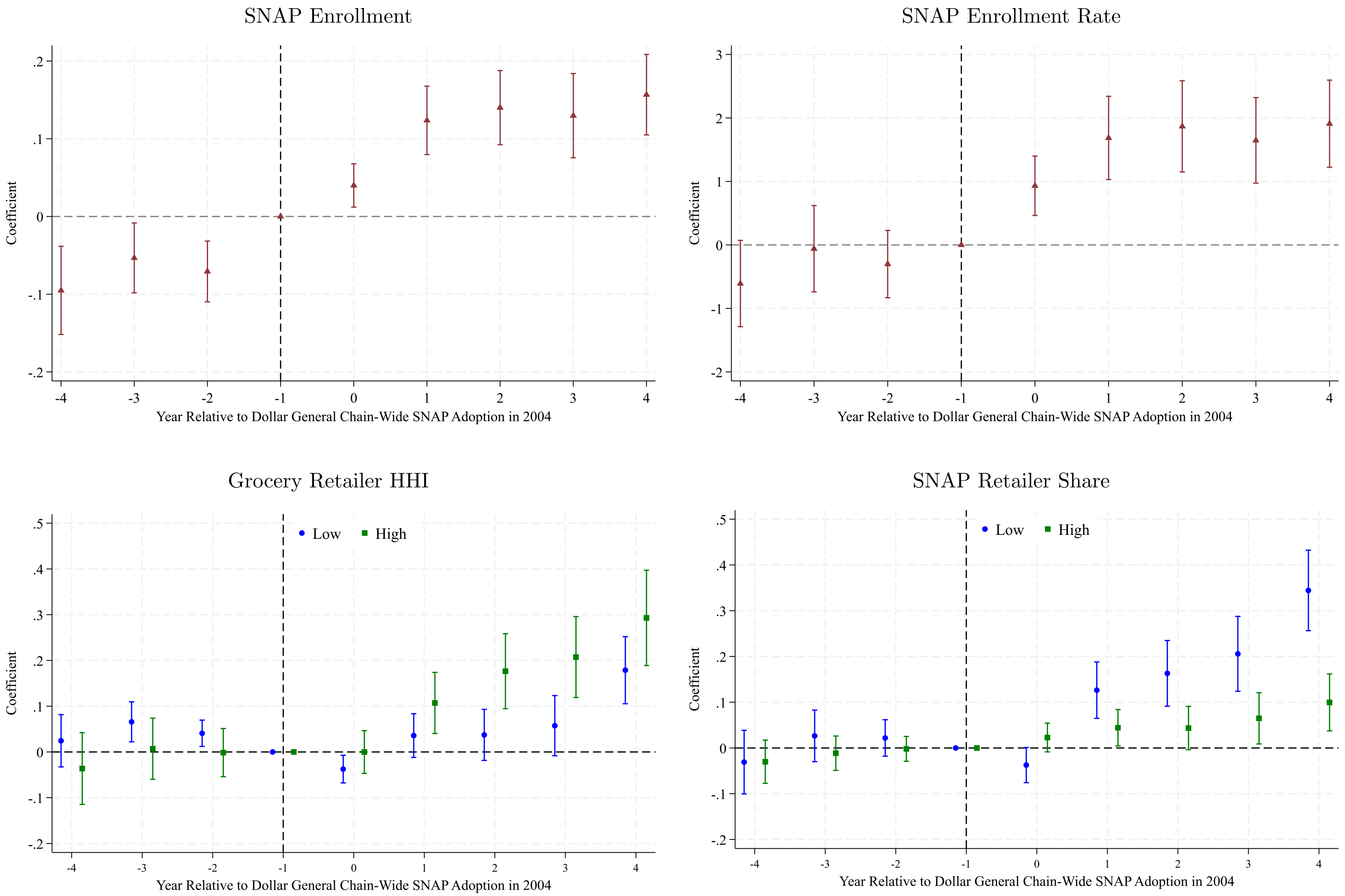

Figure 3: Event Study Estimates for Mechanisms Underlying SNAP Retailer Spillovers

Notes: This figure visualizes the dynamic effects corresponding to the mechanisms reported in Table 3. In the bottom two panels, we present results that compare the bottom tercile (low group) with the upper tercile (high group), though the full model includes all three terciles. Circle (blue) denotes the estimates for the low group, and square (green) denotes the estimates for the high group. The dependent variables in the upper panels are the log of the number of enrolled SNAP individuals and SNAP enrollment per capita, respectively. The dependent variable in the bottom panels is the log(count + 1) of SNAP-authorized grocery stores (excluding Dollar General). The x-axis denotes years relative to 2004, the year of Dollar General's chain-wide SNAP adoption. Each point is a coefficient estimate; bars show 95% confidence intervals.

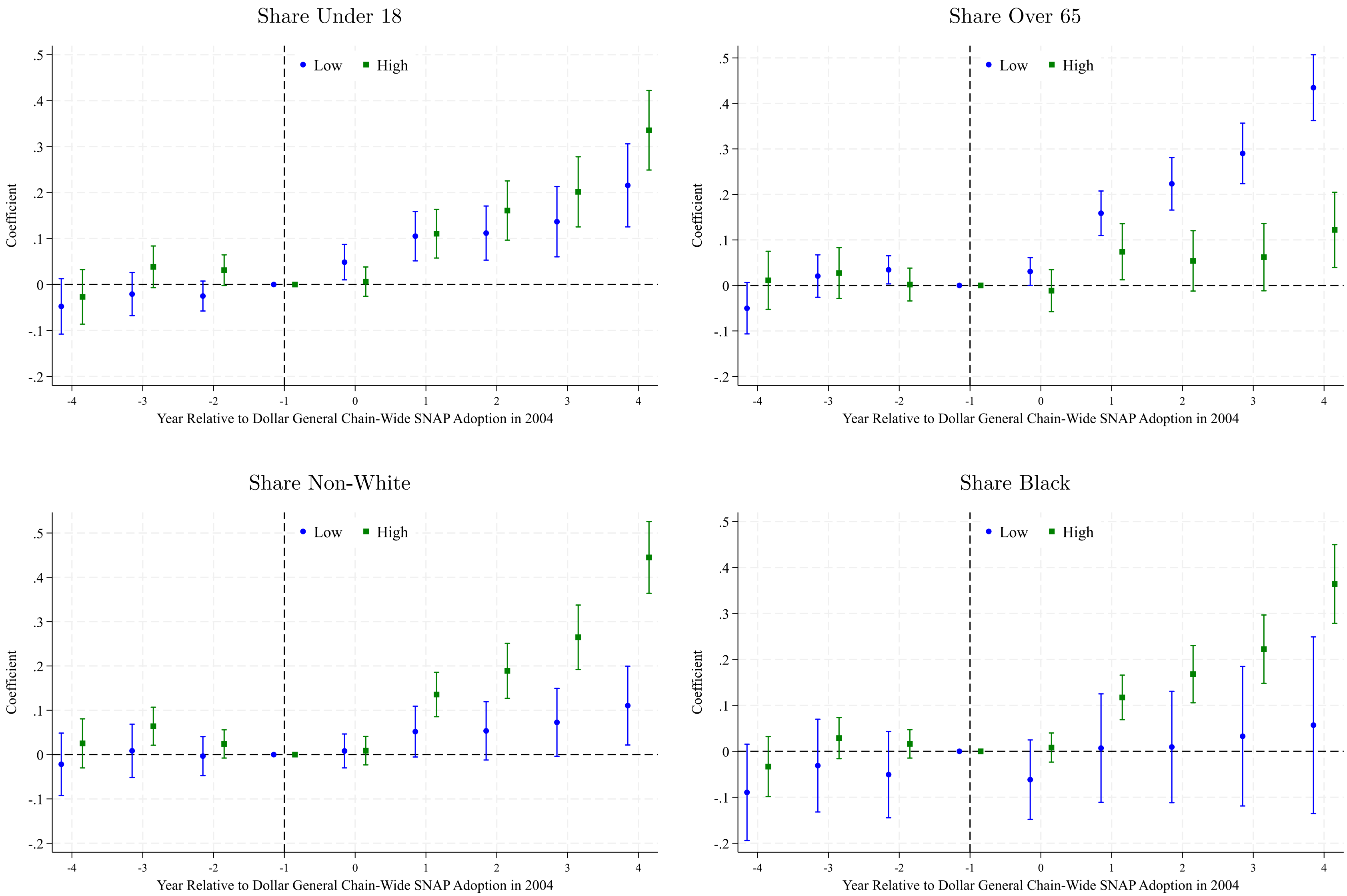

Figure 4: Event Study of Dollar General's SNAP Adoption by County Demographics

Notes: This figure presents event study estimates of Dollar General's SNAP adoption by county demographics in 2000, based on the specification in Equation (4). Each panel corresponds to four demographic variables reported in Table 4. We present results that compare the bottom tercile (low group) with the upper tercile (high group), though the full model includes all three terciles. Circle (blue) denotes the estimates for the low group, and square (green) denotes the estimates for the high group. The dependent variable is the log(count + 1) of SNAP-authorized grocery stores (excluding Dollar General). The x-axis denotes years relative to 2004, the year of Dollar General's chain-wide SNAP adoption. Each point represents the estimated coefficient for the corresponding period; bars represent 95% confidence intervals.

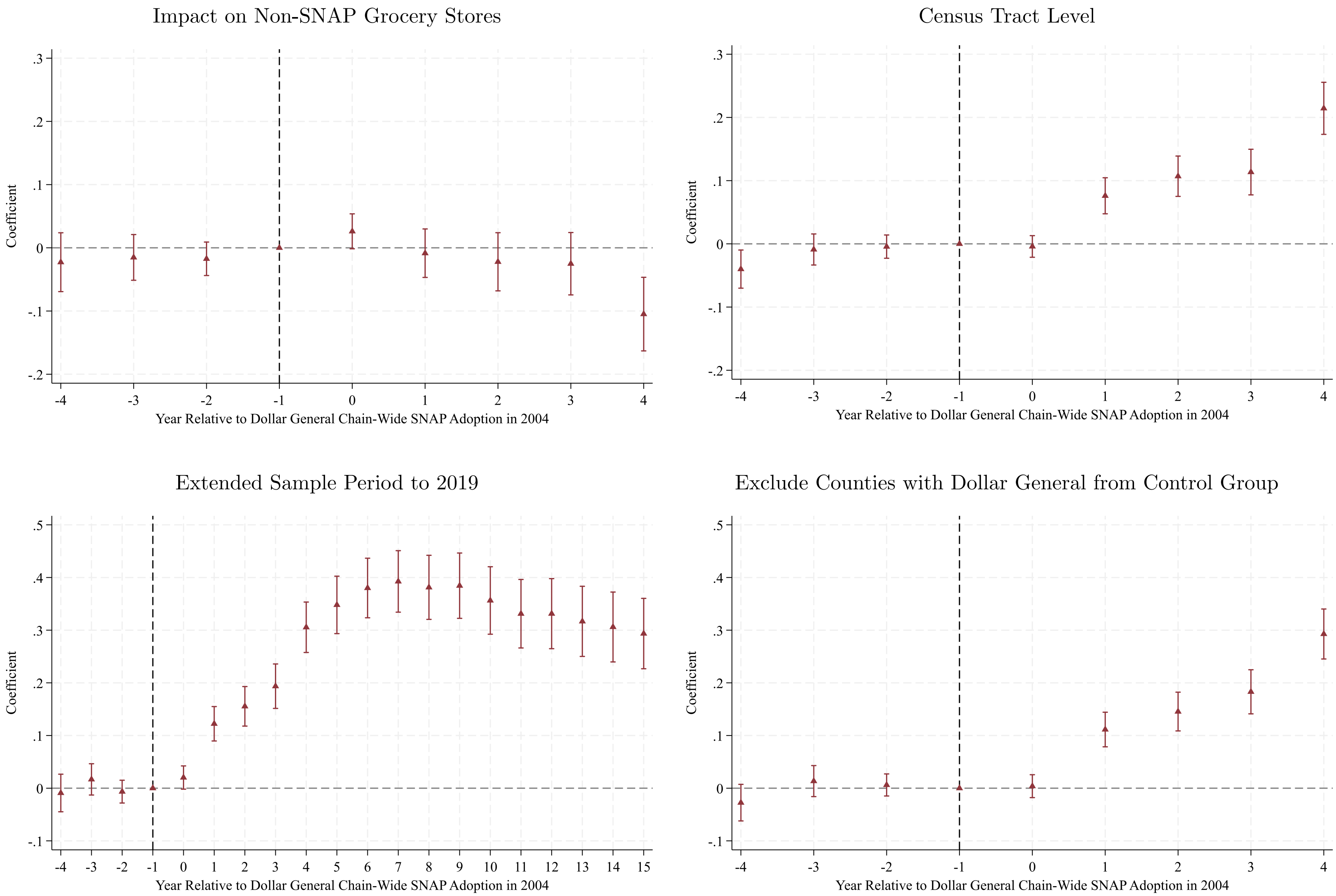

Figure 5: Event Study Robustness Checks of Dollar General's SNAP Adoption Effects

Notes: This figure presents event study estimates of Dollar General's SNAP adoption effects from robustness check models. Each panel corresponds to a different robustness specification reported in Table 5. The dependent variable is the log(count + 1) of SNAP-authorized grocery stores, except in panel A, which uses the log(count + 1) of non-SNAP-authorized grocery stores (excluding Dollar General). The x-axis denotes years relative to 2004, the year of Dollar General's chain-wide SNAP adoption. Each point represents the estimated coefficient for the corresponding period; bars represent 95% confidence intervals.

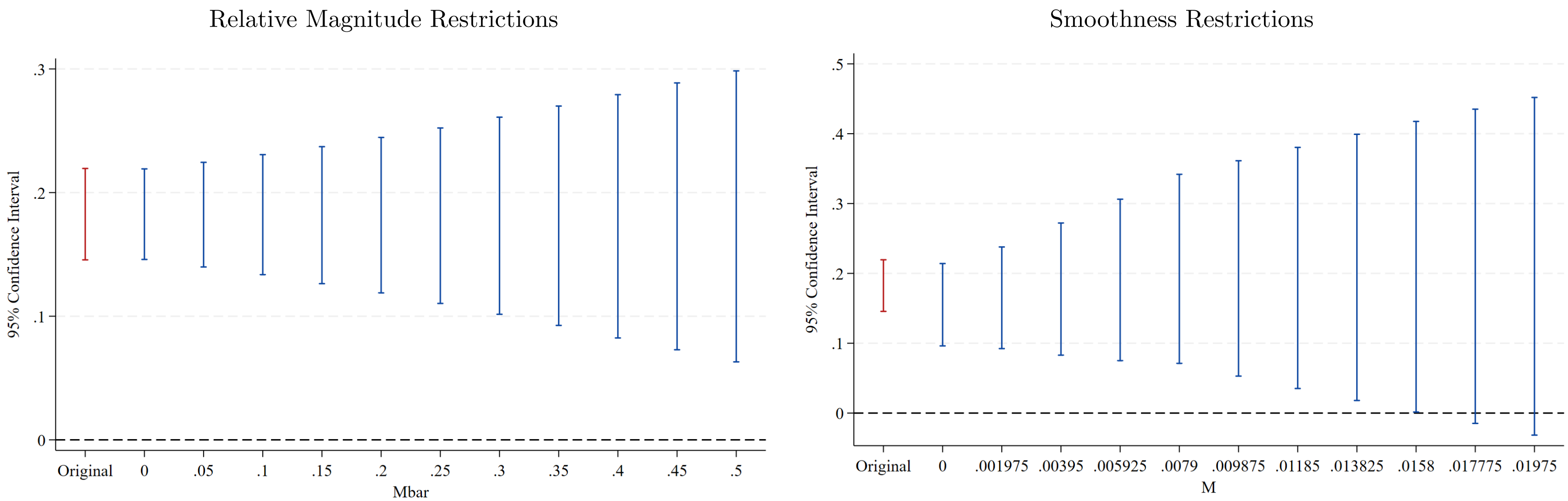

Figure 6: Sensitivity Analysis of Event Study Estimates Using Honest DiD

Notes: This figure reports sensitivity analyses based on (Rambachan and Roth, 2023) using the Honest DiD framework. The figure on the left panel presents bounds on the average treatment effect for the post-treatment period (post_1 to post_4) under Relative Magnitude Restrictions. The figure on the right panel reports the bounds under Smoothness Restrictions. Each panel shows how much deviation from parallel pre-treatment trends would be required to render the estimated effects statistically insignificant.

Appendix

Table A.1: Estimates for Table 2 using Linear Time Trend Specification

	SNAP Retailer Share (1)	SNAP Retailer Density (2)	SNAP Retailer Density (Low-Income) (3)
Treated × Post	6.363***	0.027***	0.159***
	(1.529)	(0.007)	(0.051)
County Fixed Effects	Y	Y	Y
Year Fixed Effects	Y	Y	Y
Controls × Linear Trend	Y	Y	Y
Observations	11,791	12,033	12,059

Notes: This table presents robustness estimates using county and year fixed effects, with control variables (measured in 2000) interacted with a first-order polynomial (linear) time trend. These results correspond to columns (6)–(8) in Table 2. The dependent variables are the share of SNAP-authorized grocery stores (column 1), SNAP retailer density per 1,000 population (column 2), and SNAP retailer density per 1,000 low-income population (column 3), all excluding SNAP-authorized Dollar General stores. Standard errors (in parentheses) are clustered at the county level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A.2: Estimates for Table 3 using Linear Time Trend Specification

	(1) SNAP Enrollment	(2) SNAP Enrollment Rate	(3) Grocery Retailer HHI	(4) SNAP Retailer Share
Treated × Post	0.172***	1.854***	0.049*	0.235***
	(0.014)	(0.145)	(0.025)	(0.029)
Treated × Post × Grocery Store HHI			0.030**	
			(0.012)	
Treated × Post × SNAP Retailer Share				-0.0004***
				(0.000)
County Fixed Effects	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y
Controls × Linear Trend	Y	Y	Y	Y
Observations	9,083	9,065	11,853	11,826

Notes: This table reports robustness estimates for Table 3 using a linear time trend specification. The dependent variables are log(count + 1) of SNAP-authorized grocery stores (excluding SNAP-authorized Dollar General stores) in columns (3) and (4), and individual SNAP participation outcomes in columns (1) and (2). Each column reports the interaction between Treated × Post and a baseline food retail market characteristic (measured in 2000) or tests for market expansion mechanisms. Grocery Retailer HHI refers to the Herfindahl-Hirschman Index measuring grocery market concentration in 2000. SNAP Retailer Share is the percentage of grocery stores that were SNAP-authorized in 2000. All models include county and year fixed effects and control variables interacted with a first-order polynomial (linear) time trend. Standard errors (in parentheses) are clustered at the county level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A.3: Estimates for Table 4 using Linear Time Trend Specification

	(1)	(2)	(3)	(4)
Treated × Post	-0.254***	0.413***	0.054**	0.093***
	(0.036)	(0.064)	(0.024)	(0.021)
Treated × Post × Share Under 18	0.002***			
	(0.000)			
Treated × Post × Share Over 65		-0.019***		
		(0.004)		
Treated × Post × Share Non-White			0.001***	
			(0.000)	
Treated × Post × Share Black				0.001***
				(0.000)
County Fixed Effects	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y
Controls × Linear Trend	Y	Y	Y	Y
Observations	12,060	12,060	12,060	12,033

Notes: This table reports robustness estimates for Table 4 using a linear time trend specification. The dependent variable is log(count + 1) of SNAP-authorized grocery stores (excluding SNAP-authorized Dollar General stores). Each column reports the interaction between Treated × Post and a demographic characteristic (measured in 2000): share under 18, share over 65, share non-White, and share Black. All regressions include county and year fixed effects and control variables interacted with a first-order polynomial (linear) time trend. Standard errors (in parentheses) are clustered at the county level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A.4: Estimates for Table 5 using Linear Time Trend Specification

	Impact on Non-SNAP Grocery Stores	Census Tract Model	Extended Sample Period to 2019	Exclude Counties with Dollar General from Control Group
	(1)	(2)	(3)	(4)
Treated × Post	-0.013	0.112***	0.271***	0.148***
	(0.020)	(0.014)	(0.025)	(0.017)
County / Tract Fixed Effects	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y
Controls × Linear Trend	Y	Y	Y	Y
Observations	11,074	337,122	25,900	12,018

Notes: This table reports robustness estimates for Table 5 using a linear time trend specification. Column (1) uses the log(count + 1) of non-SNAP-authorized grocery stores as the dependent variable. Column (2) re-estimates the model at the census tract level. Column (3) extends the sample period through 2019. Column (4) excludes control counties where Dollar General was present. All models include county or tract fixed effects, year fixed effects, and control variables (measured in 2000) interacted with a first-order polynomial (linear) time trend. Standard errors (in parentheses) are clustered at the corresponding geographic level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.